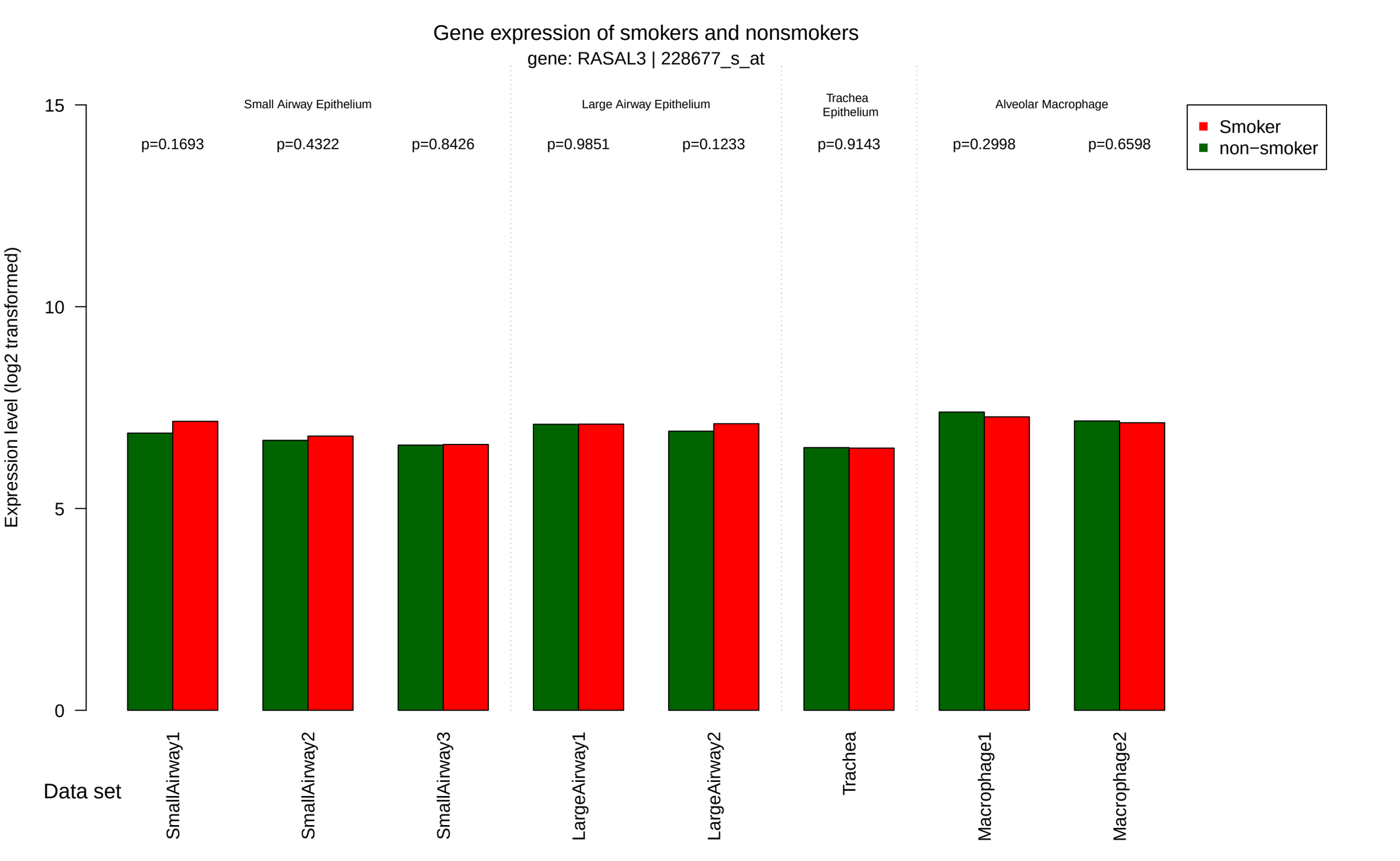
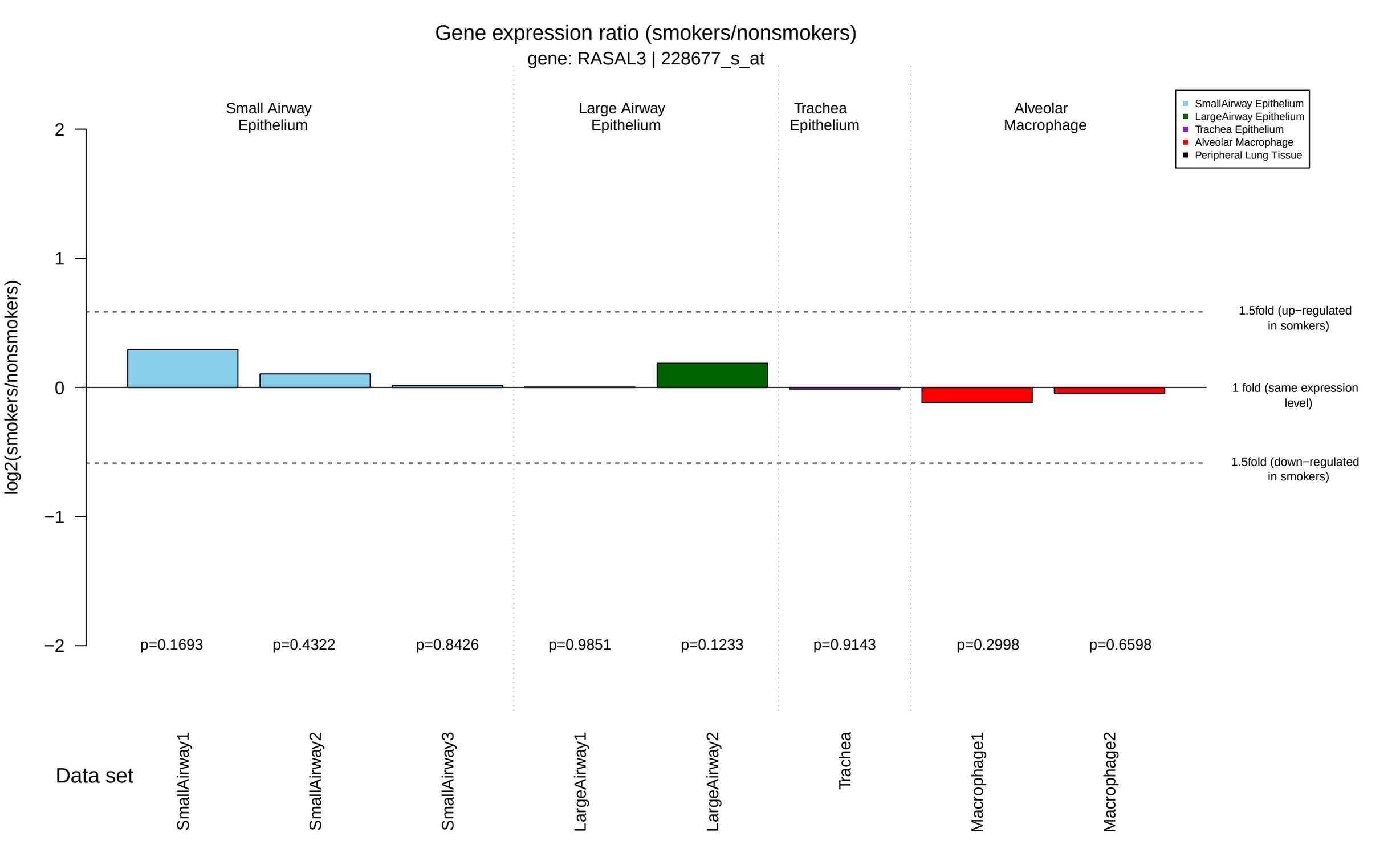
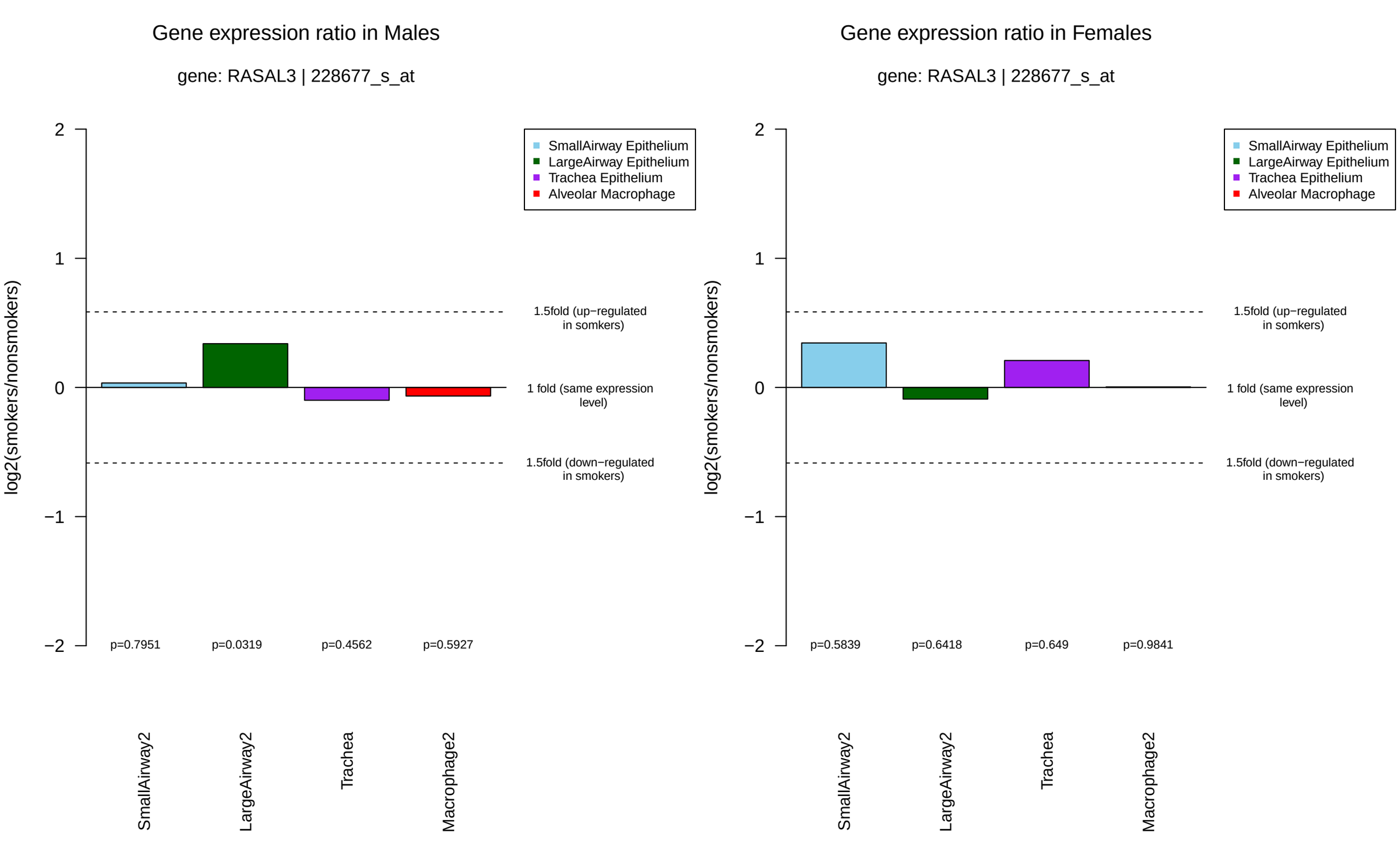
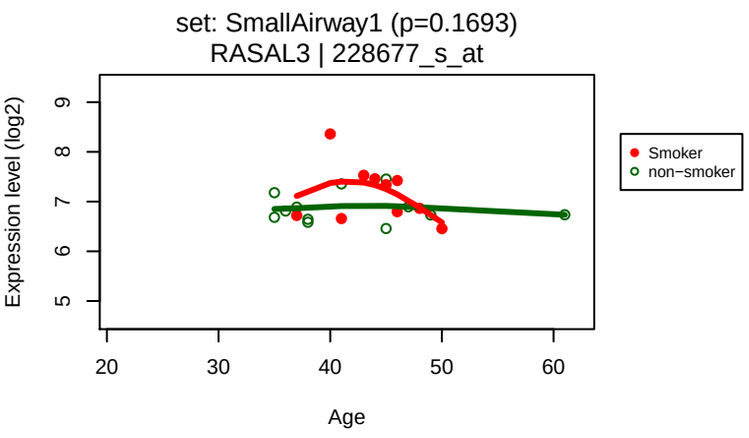
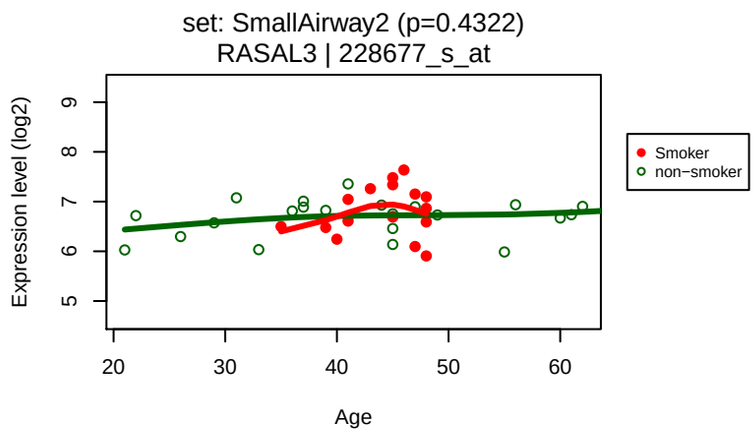
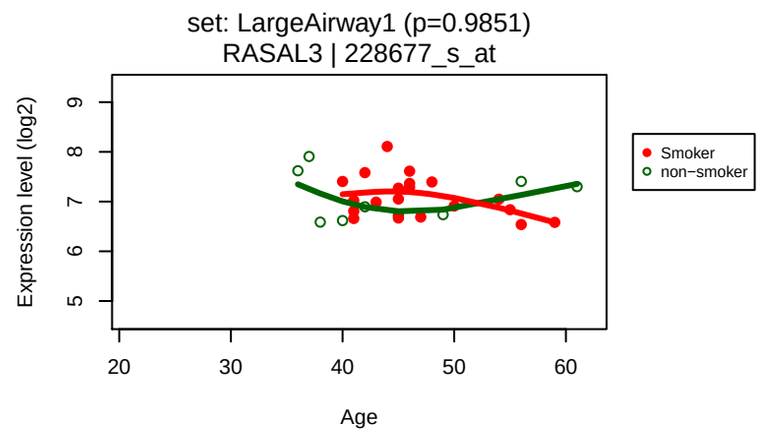

Gene expression of smokers and nonsmokers
gene: RASAL3 | 228677_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1693
p=0.4322
p=0.8426
p=0.9851
p=0.1233
p=0.9143
p=0.2998
p=0.6598
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

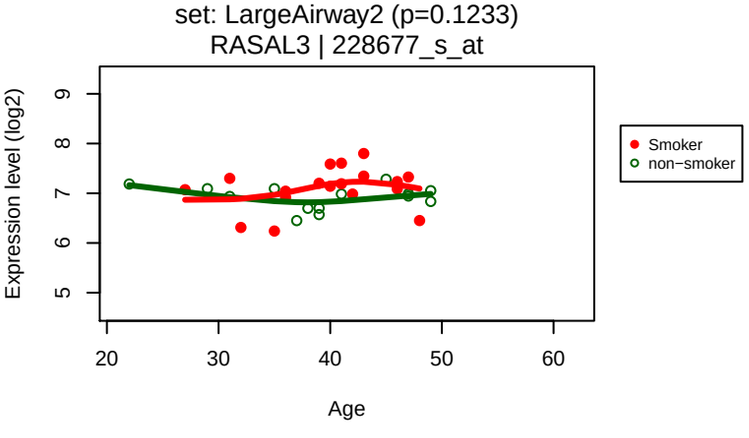

Gene expression ratio (smokers/nonsmokers)
gene: RASAL3 | 228677_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1693
p=0.4322
p=0.8426
p=0.9851
p=0.1233
p=0.9143
p=0.2998
p=0.6598
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

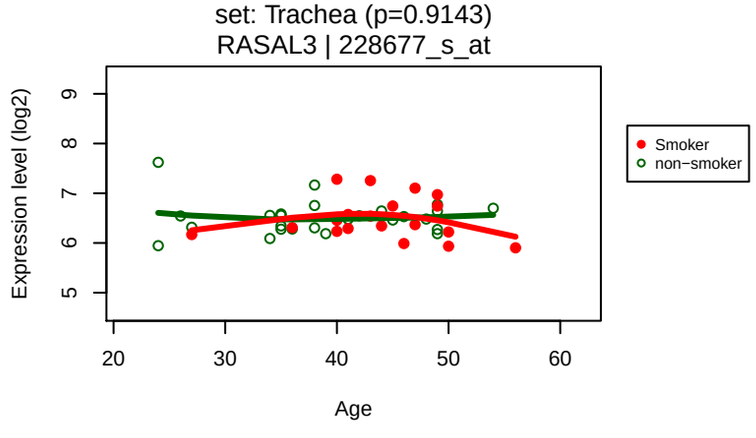

Gene expression ratio in Males
gene: RASAL3 | 228677_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7951
p=0.0319
p=0.4562
p=0.5927
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RASAL3 | 228677_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5839
p=0.6418
p=0.649
p=0.9841
SmallAirway2
LargeAirway2
Trachea
Macrophage2

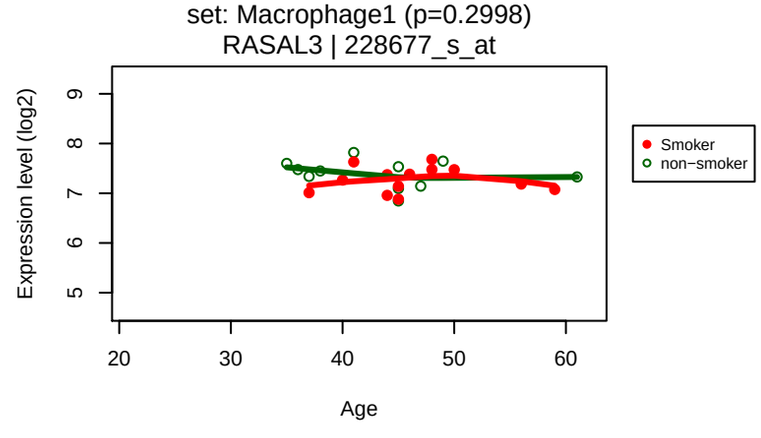

set: SmallAirway1 (p=0.1693)
RASAL3 | 228677_s_at
Expression level (log2)
Age
Smoker
non−smoker

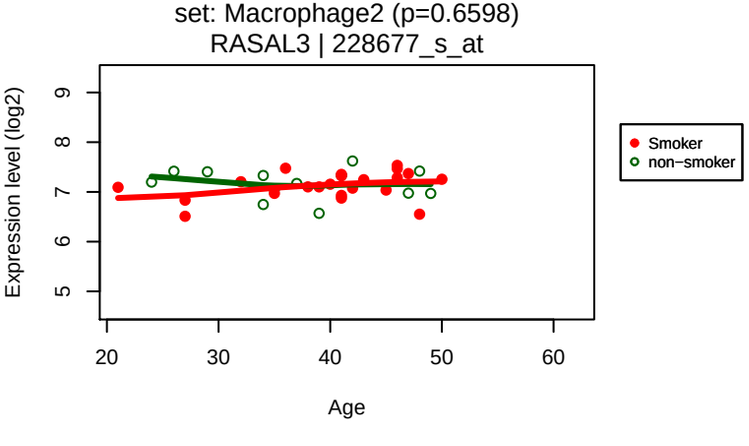

set: SmallAirway2 (p=0.4322)
RASAL3 | 228677_s_at
Expression level (log2)
Age
Smoker
non−smoker

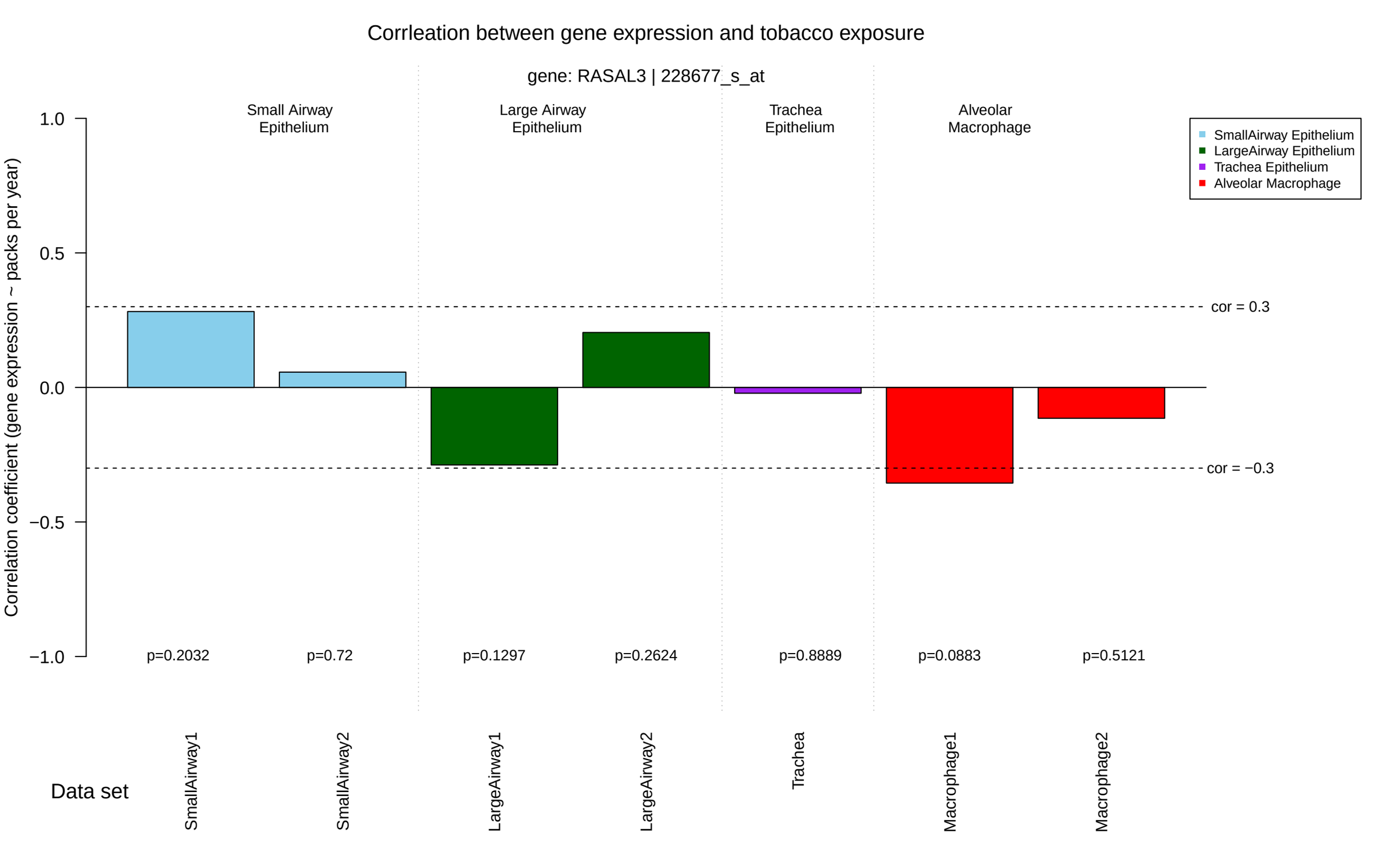

set: LargeAirway1 (p=0.9851)
RASAL3 | 228677_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1233)
RASAL3 | 228677_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.9143)
RASAL3 | 228677_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2998)
RASAL3 | 228677_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6598)
RASAL3 | 228677_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RASAL3 | 228677_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2032
p=0.72
p=0.1297
p=0.2624
p=0.8889
p=0.0883
p=0.5121
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2